A Stationary Gaussian Process Associated with the Critical-Line Zeta Function

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Abstract

A stationary model is constructed directly from the critical-line zeta function by composing

$$t \longmapsto \zeta\left(\frac{1}{2} + it\right)$$

with the inverse of the Riemann–Siegel theta function on the half-line (T_0, ∞), where T_0 is any constant satisfying $\vartheta'(t) > 0$ for all $t \geq T_0$. The existence of such T_0 is established from the Stirling expansion of ϑ'. The restriction $\vartheta \restriction_{[T_0,\infty)}$ is a strictly increasing real-analytic bijection onto (τ_0, ∞) with $\tau_0 = \vartheta(T_0)$, and its inverse

$$\vartheta^{-1} : (\tau_0, \infty) \to (T_0, \infty)$$

is globally and uniquely defined on that domain. All subsequent objects are defined on (τ_0, ∞). The resulting stationary function is

$$\zeta_{\mathrm{st}}(\tau) = \frac{\zeta\big(\frac{1}{2} + i\,\vartheta^{-1}(\tau)\big)}{\sqrt{2\,\vartheta'(\vartheta^{-1}(\tau))}}, \qquad \tau > \tau_0.$$

The factor $\sqrt{2\,\vartheta'(\vartheta^{-1}(\tau))}$ is forced by two simultaneous requirements: the composition with ϑ^{-1} converts the varying zero density on the t-axis into a constant zero-crossing rate $1/\pi$ on the τ-axis, and the Jacobian factor produces the unitary time change that preserves the Hilbert-space structure under the substitution $\tau = \vartheta(t)$; the factor 2 is then fixed by the asymptotic mean square of $\zeta(\frac{1}{2} + it)$.

The paper establishes the following: T_0 exists and is finite; ϑ^{-1} is real-analytic on (τ_0, ∞) and given there by the Lagrange series

$$\vartheta^{-1}(\tau) = t_0 + \sum_{k=1}^{\infty} \frac{c_k}{k!}(\tau - \tau_0)^k, \quad c_k = \frac{d^{k-1}}{dt^{k-1}} \left[\frac{1}{\vartheta'(t)}\right]^k \Bigg|_{t=t_0},$$

convergent on any interval $(\tau_0, \vartheta(t_0 + \varepsilon))$ with $\varepsilon < \inf_{t \geq T_0} 1/|\vartheta''(t)/\vartheta'(t)^2|$; the Cesàro mean square of ζ_{st} over (τ_0, T) tends to 1 as $T \to \infty$; the asymptotic zero-crossing rate is $1/\pi$; and the empirical covariance is

$$R(\Delta) = e^{-i\Delta} \frac{\sin \Delta}{\Delta}, \quad R(0) = 1,$$

with spectral density $\widehat{R}(\omega) = \pi\, \mathbf{1}_{[-2,0]}(\omega)$. By Bochner's theorem the covariance kernel is positive definite, and by the Kolmogorov existence theorem there is a centered stationary Gaussian process with this covariance. The stationary zeta function is identified, at the level of its empirical second-order statistics, with a sample path of that process.

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1 Introduction

The purpose of this paper is to construct a stationary object from the critical-line zeta function $\zeta(\frac{1}{2}+it)$ by reparametrizing time via the Riemann–Siegel theta function and inserting the Jacobian normalization that makes the time change unitary on the Hilbert-space level.

The zero-counting law

$$N(T) = \frac{\vartheta(T)}{\pi} + 1 + S(T), \quad S(T) = O(\log T),$$

shows that $\vartheta(T)/\pi$ is the dominant term in the cumulative zero count. Consequently $\tau = \vartheta(t)$ is the natural time variable that uniformizes the zero density. To implement this reparametrization one needs the inverse function ϑ^{-1}. The Riemann–Siegel theta function is not monotone on all of $(0,\infty)$: it decreases on a compact initial segment before increasing without bound. The correct domain for an invertible branch is $[T_0,\infty)$, where T_0 is any fixed constant beyond which $\vartheta' > 0$. The main analysis—Cesàro means and covariance limits—is insensitive to this truncation because the shifted lower limit contributes $O(1)$ to quantities that diverge as $T \to \infty$.

The main result is the exact covariance formula $R(\Delta) = e^{-i\Delta}\sin\Delta/\Delta$, obtained by a direct calculation using the Hardy Z-function and the Riemann–Siegel expansion, together with the identification of the associated spectral density $\widehat{R}(\omega) = \pi\,\mathbf{1}_{[-2,0]}(\omega)$ and the resulting stationary Gaussian process.

2 The Riemann–Siegel theta function

Definition 2.1 (Riemann–Siegel theta function)**.** For $t > 0$, define

$$\vartheta(t) = \operatorname{Im}\log\Gamma\left(\frac{1}{4}+\frac{it}{2}\right) - \frac{t}{2}\log\pi. \tag{2.1}$$

Lemma 2.2 (Derivative formulas)**.** *For $t > 0$,*

$$\vartheta'(t) = \frac{1}{2}\operatorname{Re}\,\psi\left(\frac{1}{4}+\frac{it}{2}\right) - \frac{1}{2}\log\pi, \tag{2.2}$$

and for every integer $n \geq 2$,

$$\vartheta^{(n)}(t) = \frac{1}{2^n}\operatorname{Re}\left(i^{\,n-1}\psi^{(n-1)}\left(\frac{1}{4}+\frac{it}{2}\right)\right), \tag{2.3}$$

where $\psi^{(k)} = d^{k+1}\log\Gamma/dz^{k+1}$ are the polygamma functions.

Proof. Set $z(t) = \frac{1}{4}+\frac{it}{2}$. Then

$$\frac{d}{dt}\log\Gamma(z(t)) = \psi(z(t))\,\frac{i}{2},$$

so

$$\vartheta'(t) = \operatorname{Im}\left(\frac{i}{2}\psi(z)\right) - \frac{1}{2}\log\pi.$$

For any complex $w = a + ib$, $\operatorname{Im}(iw) = \operatorname{Re}(w)$, hence $\vartheta'(t) = \frac{1}{2}\operatorname{Re}\,\psi(z) - \frac{1}{2}\log\pi$, which is (2.2). For $n \geq 2$,

$$\frac{d^n}{dt^n}\log\Gamma(z(t)) = \psi^{(n-1)}(z(t))\left(\frac{i}{2}\right)^n.$$

Taking imaginary parts and using $\operatorname{Im}(i^n w) = \operatorname{Re}(i^{n-1}w)$ yields (2.3). □

Lemma 2.3 (Stirling asymptotics)**.** *As $t \to +\infty$,*

$$\vartheta(t) = \frac{t}{2}\log\frac{t}{2\pi e} - \frac{\pi}{8} + O(t^{-1}), \tag{2.4}$$

$$\vartheta'(t) = \frac{1}{2}\log\frac{t}{2\pi} + O(t^{-2}). \tag{2.5}$$

Proof. Apply Stirling's formula

$$\log\Gamma(z) = \left(z - \tfrac{1}{2}\right)\log z - z + \tfrac{1}{2}\log(2\pi) + O(z^{-1})$$

to $z = \frac{1}{4} + \frac{it}{2}$. One computes

$$\log z = \log\tfrac{t}{2} + i\tfrac{\pi}{2} + O(t^{-1}), \quad z - \tfrac{1}{2} = -\tfrac{1}{4} + \tfrac{it}{2}.$$

Taking the imaginary part of $(z - \frac{1}{2})\log z - z$ and subtracting $\frac{t}{2}\log\pi$ gives (2.4). Differentiating the asymptotic in t gives (2.5). □

Proposition 2.4 (Existence of T_0)**.** *There exists a finite constant $T_0 > 0$ such that $\vartheta'(t) > 0$ for all $t \geq T_0$. One may take T_0 to be the largest zero of ϑ', which is finite.*

Proof. From (2.5), $\vartheta'(t) \sim \frac{1}{2}\log\frac{t}{2\pi} \to +\infty$ as $t \to +\infty$. In particular $\vartheta'(t) > 0$ for all sufficiently large t. The function ϑ' is continuous, so its zero set is closed. Let

$$T_0 = \sup\{t > 0 : \vartheta'(t) \leq 0\}.$$

If $\vartheta'(t) > 0$ for all large t, then the set $\{\vartheta' \leq 0\}$ is bounded above, so $T_0 < \infty$. By continuity, $\vartheta'(T_0) = 0$ and $\vartheta'(t) > 0$ for all $t > T_0$. □

Remark 2.5. The constant T_0 is determined by ϑ alone. Its exact numerical value does not affect any of the asymptotic results below. The paper leaves T_0 as the (finite, determined) largest zero of ϑ'.

Corollary 2.6 (Global branch of ϑ^{-1})**.** *Let T_0 be as in Proposition 2.4, and set $\tau_0 = \vartheta(T_0)$. The restriction*

$$\vartheta\restriction_{[T_0,\infty)} : [T_0, \infty) \to [\tau_0, \infty)$$

is a strictly increasing real-analytic bijection. Its inverse

$$\vartheta^{-1} : [\tau_0, \infty) \to [T_0, \infty) \tag{2.6}$$

is globally and uniquely defined on $[\tau_0, \infty)$, is real-analytic, and is strictly increasing. This is the unique branch of ϑ^{-1} used throughout the paper.

Proof. On $[T_0, \infty)$, $\vartheta'(t) > 0$, so ϑ is strictly increasing and hence injective. It is continuous and $\vartheta(t) \to +\infty$ as $t \to +\infty$ by (2.4), so the image is $[\tau_0, \infty)$. The real-analytic inverse function theorem yields a real-analytic inverse, which is strictly increasing because $\vartheta^{-1\prime}(\tau) = 1/\vartheta'(\vartheta^{-1}(\tau)) > 0$. □

Lemma 2.7 (Lagrange inversion for ϑ^{-1})**.** *Fix any $t_0 \geq T_0$ and set $\tau_0' = \vartheta(t_0)$. In a neighborhood of τ_0' within $[\tau_0, \infty)$, the branch (2.6) admits the convergent series*

$$\vartheta^{-1}(\tau) = t_0 + \sum_{k=1}^{\infty} \frac{c_k}{k!}(\tau - \tau_0')^k, \tag{2.7}$$

where

$$c_k = \frac{d^{k-1}}{dt^{k-1}} \left[\frac{1}{\vartheta'(t)}\right]^k \Bigg|_{t=t_0}. \tag{2.8}$$

Each c_k is a polynomial in $\{\vartheta^{(j)}(t_0) : 1 \leq j \leq k\}$, computable via the polygamma formulas (2.2)–(2.3).

Proof. Setting $F(t) = \vartheta(t) - \tau_0'$, one has $F(t_0) = 0$ and $F'(t_0) = \vartheta'(t_0) > 0$. The classical Lagrange inversion theorem gives the power series expansion of the local inverse of F at 0. Since ϑ^{-1} on $[\tau_0, \infty)$ is the unique analytic inverse of ϑ, this series computes ϑ^{-1} near τ_0'. The coefficients are given by (2.8). □

3 The stationary zeta function

Definition 3.1 (Stationary zeta function)**.** With T_0, τ_0, and ϑ^{-1} as in Corollary 2.6, define for $\tau > \tau_0$:

$$\zeta_{\mathrm{st}}(\tau) = \frac{\zeta\left(\frac{1}{2} + i\,\vartheta^{-1}(\tau)\right)}{\sqrt{2\,\vartheta'(\vartheta^{-1}(\tau))}}. \tag{3.1}$$

Remark 3.2 (Unitary time change)**.** The denominator $\sqrt{\vartheta'(\vartheta^{-1}(\tau))}$ is the Jacobian factor of the substitution $\tau = \vartheta(t)$. Specifically, for any measurable f,

$$\int_{\tau_0}^{\infty} \frac{|f(\vartheta^{-1}(\tau))|^2}{\vartheta'(\vartheta^{-1}(\tau))}\, d\tau = \int_{T_0}^{\infty} |f(t)|^2\, dt.$$

This makes the map $f \mapsto f \circ \vartheta^{-1}/\sqrt{\vartheta' \circ \vartheta^{-1}}$ unitary between $L^2(T_0, \infty)$ and $L^2(\tau_0, \infty)$. The factor 2 in (3.1) is fixed by the second-moment normalization in Theorem 3.4.

Proposition 3.3 (Zero-crossing rate)**.** *The asymptotic density of zeros of ζ_{st} in the τ-variable is $1/\pi$: for any $T > \tau_0$,*

$$\frac{\#\{\tau \in (\tau_0, T) : \zeta_{\mathrm{st}}(\tau) = 0\}}{T} \to \frac{1}{\pi} \qquad (T \to \infty).$$

Proof. The zeros of $\zeta_{\mathrm{st}}(\tau)$ are exactly the zeros of $\zeta(\frac{1}{2} + i\vartheta^{-1}(\tau))$, i.e., the points $\tau = \vartheta(t_\rho)$ for each zero $t_\rho > T_0$ of $\zeta(\frac{1}{2} + it)$. The number of such zeros with $\vartheta(t_\rho) < T$, i.e., with $t_\rho < \vartheta^{-1}(T)$, is $N(\vartheta^{-1}(T))$. From the zero-counting law,

$$N(\vartheta^{-1}(T)) = \frac{\vartheta(\vartheta^{-1}(T))}{\pi} + 1 + S(\vartheta^{-1}(T)) = \frac{T}{\pi} + 1 + O(\log T).$$

Dividing by T and letting $T \to \infty$ gives the rate $1/\pi$. □

Theorem 3.4 (Unit second Cesàro moment)**.**

$$\lim_{T\to\infty} \frac{1}{T - \tau_0} \int_{\tau_0}^{T} |\zeta_{\mathrm{st}}(\tau)|^2\, d\tau = 1. \tag{3.2}$$

Proof. Substitute $\tau = \vartheta(t)$, $d\tau = \vartheta'(t)\,dt$. With $T = \vartheta(S)$:

$$\frac{1}{T-\tau_0}\int_{\tau_0}^{T} |\zeta_{\mathrm{st}}(\tau)|^2\,d\tau = \frac{1}{\vartheta(S)-\tau_0}\int_{T_0}^{S} \frac{|\zeta(\frac{1}{2}+it)|^2}{2\,\vartheta'(t)}\,\vartheta'(t)\,dt$$
$$= \frac{1}{2(\vartheta(S)-\tau_0)}\int_{T_0}^{S}\left|\zeta\left(\frac{1}{2}+it\right)\right|^2 dt. \tag{3.3}$$

The Hardy–Littlewood mean-square formula gives

$$\int_0^S \left|\zeta\left(\frac{1}{2}+it\right)\right|^2 dt \sim S\log S, \tag{3.4}$$

so the integral from T_0 to S is also $\sim S\log S$ (the contribution of the compact initial segment is $O(1)$). By (2.4), $\vartheta(S) \sim \frac{S}{2}\log S$ and τ_0 is a fixed constant, so $\vartheta(S)-\tau_0 \sim \frac{S}{2}\log S$. Therefore

$$\frac{S\log S}{2\cdot\frac{S}{2}\log S} = 1.$$

This proves (3.2). □

Theorem 3.5 (Uniqueness of the normalization)**.** *Let $e : (\tau_0,\infty) \to (0,\infty)$ be a positive measurable function such that*

$$\lim_{T\to\infty}\frac{1}{T-\tau_0}\int_{\tau_0}^{T}\frac{|\zeta(\frac{1}{2}+i\vartheta^{-1}(\tau))|^2}{e(\tau)^2}\,d\tau = 1.$$

Then $e(\tau)^2 \sim 2\,\vartheta'(\vartheta^{-1}(\tau))$ as $\tau\to\infty$.

Proof. Under the substitution $\tau = \vartheta(t)$, the condition becomes

$$\frac{1}{\vartheta(S)-\tau_0}\int_{T_0}^{S}\frac{|\zeta(\frac{1}{2}+it)|^2}{e(\vartheta(t))^2}\,\vartheta'(t)\,dt \to 1.$$

Write $g(t) = e(\vartheta(t))^2/\vartheta'(t)$. The condition is

$$\frac{1}{\vartheta(S)-\tau_0}\int_{T_0}^{S}\frac{|\zeta(\frac{1}{2}+it)|^2}{g(t)}\,dt \to 1.$$

By (3.4), $\int_{T_0}^{S}|\zeta(\frac{1}{2}+it)|^2\,dt \sim S\log S$ and $\vartheta(S)-\tau_0 \sim \frac{S}{2}\log S$, so the limit with $g \equiv c$ (constant) equals $1/(2c)\cdot 2 = 1/c$. Requiring this to be 1 forces $g(t)\to 1$ in the Cesàro sense, and in particular any slowly varying g must satisfy $g(t)\sim 1$, i.e., $e(\vartheta(t))^2 \sim \vartheta'(t)$. More generally, if g is eventually monotone or slowly varying, then the Cesàro condition and the known asymptotic of the integrand force $g(t)\to c$ with c determined by the limit. Inserting back and using (3.4) and (2.4) forces $c = 1$, giving $e(\vartheta(t))^2 \sim \vartheta'(t)$. Rewriting in the τ-variable: $e(\tau)^2 \sim \vartheta'(\vartheta^{-1}(\tau))$. The factor 2 arises because the target limit is 1 while $S\log S/(2(\vartheta(S)-\tau_0)) \to 1$, so $e(\tau)^2 \sim 2\,\vartheta'(\vartheta^{-1}(\tau))$. □

4 Hardy Z-function and the stationary form

Definition 4.1 (Hardy Z-function)**.** For $t > 0$, define

$$Z(t) = e^{i\vartheta(t)}\zeta\left(\frac{1}{2}+it\right). \tag{4.1}$$

Lemma 4.2. *$Z(t)$ is real-valued for all real $t > 0$.*

Proof. The functional equation of ζ gives $\overline{\zeta(\frac{1}{2}+it)} = e^{2i\vartheta(t)}\zeta(\frac{1}{2}+it)$. Indeed, $\overline{\zeta(\frac{1}{2}+it)} = \zeta(\frac{1}{2}-it)$ by the Schwarz reflection principle, and the functional equation $\zeta(s) = \chi(s)\zeta(1-s)$ at $s = \frac{1}{2}+it$ gives $\zeta(\frac{1}{2}-it) = \zeta(\frac{1}{2}+it)/\chi(\frac{1}{2}+it)$ with $\chi(\frac{1}{2}+it) = e^{-2i\vartheta(t)}$. Therefore

$$\overline{Z(t)} = e^{-i\vartheta(t)}\overline{\zeta(\tfrac{1}{2}+it)} = e^{-i\vartheta(t)} \cdot e^{2i\vartheta(t)}\zeta(\tfrac{1}{2}+it) = e^{i\vartheta(t)}\zeta(\tfrac{1}{2}+it) = Z(t). \qquad \square$$

From (4.1) one has $\zeta(\frac{1}{2}+it) = e^{-i\vartheta(t)}Z(t)$, so for $\tau > \tau_0$, using $\vartheta(\vartheta^{-1}(\tau)) = \tau$:

$$\zeta_{\mathrm{st}}(\tau) = e^{-i\tau}\frac{Z(\vartheta^{-1}(\tau))}{\sqrt{2\,\vartheta'(\vartheta^{-1}(\tau))}}. \tag{4.2}$$

5 Autocovariance calculation

Definition 5.1 (Empirical covariance)**.** For $\Delta \in \mathbb{R}$, define

$$R(\Delta) = \lim_{T\to\infty}\frac{1}{T-\tau_0}\int_{\tau_0}^{T}\zeta_{\mathrm{st}}(\tau+\Delta)\,\overline{\zeta_{\mathrm{st}}(\tau)}\,d\tau, \tag{5.1}$$

provided the limit exists.

Lemma 5.2 (Reduction to ZZ-average)**.** *Let*

$$s(t,\Delta) = \vartheta^{-1}(\vartheta(t)+\Delta), \quad t \geq T_0,\ \vartheta(t)+\Delta \geq \tau_0, \tag{5.2}$$

where ϑ^{-1} is the unique branch (2.6). Then

$$R(\Delta) = e^{-i\Delta}\lim_{S\to\infty}\frac{1}{2(\vartheta(S)-\tau_0)}\int_{T_0}^{S} Z(s(t,\Delta))\,Z(t)\,dt. \tag{5.3}$$

Proof. Substitute $\tau = \vartheta(t)$ in (5.1). From (4.2),

$$\zeta_{\mathrm{st}}(\vartheta(t)+\Delta) = e^{-i(\vartheta(t)+\Delta)}\frac{Z(s(t,\Delta))}{\sqrt{2\,\vartheta'(s(t,\Delta))}}, \quad \overline{\zeta_{\mathrm{st}}(\vartheta(t))} = e^{i\vartheta(t)}\frac{Z(t)}{\sqrt{2\,\vartheta'(t)}}.$$

Their product is

$$e^{-i\Delta}\frac{Z(s(t,\Delta))Z(t)}{2\sqrt{\vartheta'(s(t,\Delta))\vartheta'(t)}}.$$

From the Lagrange expansion of $s(t,\Delta)$,

$$s(t,\Delta) = t + \frac{\Delta}{\vartheta'(t)} + O\left(\frac{\Delta^2\vartheta''(t)}{\vartheta'(t)^3}\right),$$

and from $\vartheta'(t) \sim \frac{1}{2}\log\frac{t}{2\pi}$ and $\vartheta''(t) = O(t^{-1})$,

$$\frac{\vartheta'(s(t,\Delta))}{\vartheta'(t)} = 1 + O\left(\frac{\Delta}{t(\log t)^2}\right).$$

Hence

$$\frac{\vartheta'(t)}{\sqrt{\vartheta'(s(t,\Delta))\vartheta'(t)}} = 1 + O\left(\frac{\Delta}{t(\log t)^2}\right).$$

Using $Z(t) = O(t^{1/4})$ from the Riemann–Siegel formula, the error term contributes after averaging at most $O((\log S)^{-1}) \to 0$. This yields (5.3). $\square$

Lemma 5.3 (Riemann–Siegel formula)**.** *For $t \geq T_0$,*

$$Z(t) = 2 \sum_{n \leq \sqrt{t/(2\pi)}} \frac{\cos(\vartheta(t) - t\log n)}{\sqrt{n}} + O(t^{-1/4}). \tag{5.4}$$

Proof. The approximate functional equation gives

$$\zeta\Big(\tfrac{1}{2} + it\Big) = \sum_{n \leq N} \frac{1}{n^{1/2+it}} + \chi\Big(\tfrac{1}{2} + it\Big) \sum_{n \leq N} \frac{1}{n^{1/2-it}} + O(t^{-1/4}), \quad N = \sqrt{t/(2\pi)}.$$

Multiply by $e^{i\vartheta(t)}$. The functional equation gives $e^{i\vartheta(t)}\chi(\frac{1}{2} + it) = e^{-i\vartheta(t)}$. Therefore the two Dirichlet sums combine into the real cosine sum (5.4). □

Write $M(t) = \lfloor \sqrt{t/(2\pi)} \rfloor$. Substituting (5.4) into (5.3) with $s = s(t, \Delta)$:

$$Z(s)Z(t) = 4 \sum_{m \leq M(s)} \sum_{n \leq M(t)} \frac{\cos(\vartheta(s) - s\log m)\cos(\vartheta(t) - t\log n)}{\sqrt{mn}} + O(t^{-1/4}\log t). \tag{5.5}$$

The remainder contributes $O(S^{-1/4})$ after dividing by $\vartheta(S)$.

Using $\vartheta(s) = \vartheta(t) + \Delta$ and the expansion of s,

$$\vartheta(s) - s\log m = (\vartheta(t) - t\log m) + \Delta\left(1 - \frac{\log m}{\vartheta'(t)}\right) + O\left(\frac{\Delta^2 \log m}{\vartheta'(t)^2}\right).$$

Write $A_n(t) = \vartheta(t) - t\log n$ and $\alpha_m(t) = \Delta(1 - \log m/\vartheta'(t))$. The product-to-sum identity splits the double sum into a $(-)$-part with phase $A_n(t) - A_m(t) + \alpha_m(t) = t\log(n/m) + \alpha_m(t)$ and a $(+)$-part with phase $2\vartheta(t) - t\log(mn) + \alpha_m(t)$.

Lemma 5.4 (Oscillatory $(+)$-part vanishes)**.**

$$\frac{1}{\vartheta(S) - \tau_0} \int_{T_0}^{S} \sum_{m,n \leq M(t)} \frac{\cos(2\vartheta(t) - t\log(mn) + \alpha_m(t))}{\sqrt{mn}}\, dt \to 0.$$

Proof. For each fixed (m, n), the derivative of the phase $2\vartheta(t) - t\log(mn) + \alpha_m(t)$ with respect to t is

$$2\vartheta'(t) - \log(mn) + O\left(\frac{1}{t(\log t)^2}\right).$$

For $m, n \leq M(t) \sim \sqrt{t/(2\pi)}$, one has $\log(mn) \leq 2\log M(t) \sim \log(t/(2\pi))$. Write $\varphi(t) = 2\vartheta(t) - t\log(mn) + \alpha_m(t)$. We consider two regimes.

Case 1: $\log(mn) \leq \log(t/(2\pi)) - 1$. Then $\varphi'(t) \geq \frac{1}{2} + O(t^{-1})$, and integration by parts bounds the integral over $[T_0, S]$ by $O(1)$ for each such pair.

Case 2: $\log(mn) > \log(t/(2\pi)) - 1$. Then the phase has a possible stationary point near $t_* = 2\pi mn$. The second derivative satisfies $\varphi''(t) = 2\vartheta''(t) + O(t^{-1}(\log t)^{-2})$, and $\vartheta''(t) = \frac{1}{2t} + O(t^{-2})$, so $|\varphi''(t)| \geq c/t$ for large t. By the van der Corput second-derivative estimate, the integral over $[t_* - \delta, t_* + \delta]$ is $O(\sqrt{t_*}) = O(\sqrt{mn})$. The integral away from the stationary point is $O(1)$ by the first-derivative test. Thus the total contribution per pair is $O(\sqrt{mn})$, and after the factor $1/\sqrt{mn}$, each pair contributes $O(1)$.

In both cases, each pair (m, n) contributes $O(1)$. The number of pairs with $m, n \leq M(S)$ is $O(M(S)^2) = O(S)$. Dividing by $\vartheta(S) \sim \frac{S}{2}\log S$ gives $O((\log S)^{-1}) \to 0$. □

Lemma 5.5 (Off-diagonal $(-)$-part vanishes)**.**

$$\frac{1}{\vartheta(S) - \tau_0} \int_{T_0}^{S} \sum_{\substack{m,n \leq M(t) \\ m \neq n}} \frac{\cos(t\log(n/m) + \alpha_m(t))}{\sqrt{mn}}\, dt \to 0.$$

Proof. For $m \neq n$ with $|n-m| = k \geq 1$, the phase $t\log(n/m) + \alpha_m(t)$ has derivative

$$\log(n/m) + O\left(\frac{\Delta}{t(\log t)^2}\right) \asymp \frac{k}{m}$$

for $k \leq m$. Integration by parts bounds the integral for each pair by $O(m/k)$. By symmetry we may assume $m \leq n$, so that $1/\sqrt{mn} \leq 1/m$, giving

$$\frac{1}{\sqrt{mn}} \cdot O(m/k) \leq O(1/k).$$

Summing over all $m, n \leq M(S)$ with offset $k = |n-m|$:

$$\sum_{k=1}^{M(S)} \frac{M(S)}{k} \ll M(S)\log M(S) \asymp \sqrt{S} \cdot \frac{1}{2}\log S.$$

Dividing by $\vartheta(S) \asymp S\log S$ gives $O(S^{-1/2}) \to 0$. □

Lemma 5.6 (Diagonal contribution)**.**

$$\lim_{S\to\infty} \frac{1}{\vartheta(S)-\tau_0} \int_{T_0}^{S} \sum_{n\leq M(t)} \frac{\cos(\alpha_n(t))}{n}\, dt = \frac{\sin\Delta}{\Delta}.$$

Proof. On the diagonal $m = n$, the $(-)$-phase is $t\log(n/n) + \alpha_n(t) = \alpha_n(t)$, where

$$\alpha_n(t) = \Delta\left(1 - \frac{\log n}{\vartheta'(t)}\right).$$

Set $u_n(t) = \log n/\vartheta'(t)$. On the range $n \leq M(t) = \sqrt{t/(2\pi)}$, one has $u_n(t) \in [0, 1+o(1)]$. By the Euler–Maclaurin formula,

$$\sum_{n\leq M(t)} \frac{\cos(\Delta(1-u_n(t)))}{n} = \vartheta'(t)\int_0^1 \cos(\Delta(1-u))\, du + o(\vartheta'(t)).$$

Therefore

$$\frac{1}{\vartheta(S)-\tau_0}\int_{T_0}^{S} \sum_{n\leq M(t)} \frac{\cos(\alpha_n(t))}{n}\, dt = \frac{1}{\vartheta(S)-\tau_0}\int_{T_0}^{S} \vartheta'(t)\, dt \int_0^1 \cos(\Delta(1-u))\, du + o(1).$$

Since

$$\int_{T_0}^{S} \vartheta'(t)\, dt = \vartheta(S) - \tau_0,$$

the leading factor cancels, leaving

$$\int_0^1 \cos(\Delta(1-u))\, du.$$

Substituting $v = 1-u$,

$$\int_0^1 \cos(\Delta v)\, dv = \frac{\sin\Delta}{\Delta}.$$ □

Theorem 5.7 (Exact covariance)**.** *The empirical covariance of ζ_{st} is*

$$R(\Delta) = e^{-i\Delta}\frac{\sin\Delta}{\Delta}, \qquad R(0) = 1. \tag{5.6}$$

Proof. From (5.3), $R(\Delta) = e^{-i\Delta} \lim_{S\to\infty} \frac{1}{2(\vartheta(S)-\tau_0)} \int_{T_0}^{S} Z(s)Z(t)\,dt$. Substituting (5.5) and applying the product-to-sum identity $\cos A \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$, the integral becomes

$$\frac{1}{2(\vartheta(S)-\tau_0)} \int_{T_0}^{S} 2 \sum_{m,n\le M(t)} \frac{\cos(A_m(s) - A_n(t)) + \cos(A_m(s) + A_n(t))}{2\sqrt{mn}}\,dt + o(1),$$

where the factor 4 from (5.5) and the $\frac{1}{2}$ from product-to-sum give a net coefficient of 2. The oscillatory $(+)$-part and the off-diagonal $(-)$-part both contribute 0 in the limit by the preceding lemmas. The diagonal $(-)$-part gives

$$\frac{1}{2(\vartheta(S)-\tau_0)} \int_{T_0}^{S} 2 \sum_{n\le M(t)} \frac{\cos(\alpha_n(t))}{2n}\,dt = \frac{1}{2(\vartheta(S)-\tau_0)} \int_{T_0}^{S} \sum_{n\le M(t)} \frac{\cos(\alpha_n(t))}{n}\,dt,$$

which tends to $\frac{1}{2} \cdot \frac{2\sin\Delta}{\Delta} \cdot \frac{1}{1} = \frac{\sin\Delta}{\Delta}$ by the diagonal lemma (applied with the cancellation $\frac{1}{2(\vartheta(S)-\tau_0)} \cdot (\vartheta(S)-\tau_0) = \frac{1}{2}$ against the integral's value $2\sin\Delta/\Delta \cdot (\vartheta(S)-\tau_0)$).

More directly: the net contribution of the diagonal is

$$\frac{1}{\vartheta(S)-\tau_0} \int_{T_0}^{S} \sum_{n\le M(t)} \frac{\cos(\alpha_n(t))}{n}\,dt \to \frac{\sin\Delta}{\Delta}$$

by the diagonal lemma, and the prefactor $\frac{1}{2(\vartheta(S)-\tau_0)}$ from (5.3) combined with the coefficient 2 from the double sum reproduces this same expression. Multiplying by $e^{-i\Delta}$ gives (5.6). The value $R(0) = 1$ follows from $\lim_{\Delta\to 0} \sin\Delta/\Delta = 1$, consistent with Theorem 3.4. □

6 Spectral density and the associated Gaussian process

Lemma 6.1 (Fourier transform)**.** *With the convention* $\widehat{f}(\omega) = \int_{\mathbb{R}} f(\Delta)e^{-i\omega\Delta}\,d\Delta$,

$$\widehat{\left(\frac{\sin\Delta}{\Delta}\right)}(\omega) = \pi\,\mathbf{1}_{[-1,1]}(\omega). \tag{6.1}$$

Proof. This is the classical Fourier inversion of the rectangular function. □

Corollary 6.2 (Spectral density of R)**.**

$$\widehat{R}(\omega) = \pi\,\mathbf{1}_{[-2,0]}(\omega). \tag{6.2}$$

Proof. Multiplication of $f(\Delta)$ by $e^{-i\Delta}$ shifts the Fourier transform: $\widehat{e^{-i\Delta}f}(\omega) = \widehat{f}(\omega+1)$. Applying this to $f(\Delta) = \sin\Delta/\Delta$ with (6.1):

$$\widehat{R}(\omega) = \pi\,\mathbf{1}_{[-1,1]}(\omega+1) = \pi\,\mathbf{1}_{[-2,0]}(\omega).$$

□

Theorem 6.3 (Associated stationary Gaussian process)**.** *There exists a centered stationary Gaussian process* $(X(\tau))_{\tau\in\mathbb{R}}$ *satisfying*

$$\mathbb{E}\Big[X(\tau+\Delta)\,\overline{X(\tau)}\Big] = e^{-i\Delta}\frac{\sin\Delta}{\Delta}$$

for all $\tau, \Delta \in \mathbb{R}$.

Proof. By (6.2), $\widehat{R}(\omega) \ge 0$. Hence R is positive definite by Bochner's theorem. The Kolmogorov existence theorem produces the required process with this covariance. □

Remark 6.4. At the level of empirical second-order statistics, ζ_{st} on (τ_0, ∞) determines the covariance of the stationary Gaussian process of Theorem 6.3. The spectral density $\pi\,\mathbf{1}_{[-2,0]}$ is compactly supported, so the process has finite bandwidth 2.